

AYMANE AIT DADS

Software Engineering Intern (AI/ML) | Machine Learning · Data Analysis · AI Research

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a 30B-parameter model with LoRA SFT - directly demonstrating the research, problem-solving, and AI/ML development skills Revvity seeks. Reduced manual network analysis time by 60% at Orange Maroc by building automated ML pipelines on 100K+ records and delivering stakeholder presentations translating model outputs into actionable maintenance recommendations. Brings hands-on experience with Python, PyTorch, Scikit-learn, data analysis, and visualization tools aligned with Revvity's cross-functional software development and AI solution implementation needs. Aims to apply rigorous ablation methodology and end-to-end ML engineering experience to help Revvity's research teams identify and deploy AI/ML solutions that accelerate health science breakthroughs.

CORE COMPETENCIES

AI/ML Solution Implementation | Software Development | Data Analysis & Visualization | Problem-Solving & Reasoning | Cross-Functional Collaboration | Research & Learning | Quality Assurance & Testing | Stakeholder Communication

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built and evaluated Random Forest and K-Means AI/ML models on **100K+ fiber-optic network records**, classifying network quality across thousands of nodes with automated feature engineering.
- Developed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for business decision-making.
- Reduced manual analysis time by **~60%** through automated feature engineering and model evaluation pipelines, demonstrating measurable software development impact on operational efficiency.
- Delivered stakeholder presentations translating complex model outputs into **clear maintenance recommendations**, communicating findings across technical and non-technical cross-functional teams.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge

Kaggle Competition · In Progress (June 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context).
- Performed **task-level error analysis** across 5 reasoning categories (unit conversion, cipher, symbolic, gravity, bit manipulation); built a diagnostic evaluation pipeline revealing that reasoning over-expansion - not incorrect reasoning - was the primary failure mode, guiding targeted data and training improvements.

PyTorch · Unsloth · Hugging Face Transformers · PEFT · vLLM

AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in the EURECOM class** private Kaggle leaderboard (0.791 AUC) by fine-tuning a CLIP ViT-L/14 backbone (428M parameters) with a custom MLP head, dual learning rates, and 10-view test-time augmentation.
- Submitted independently to the **NTIRE 2026 @ CVPR CodaBench** international benchmark; discovered that 1-epoch training (0.793 AUC) outperforms 4-epoch (0.767 AUC), demonstrating out-of-distribution generalization through controlled ablation experiments.

PyTorch · OpenCLIP · Scikit-learn · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Statistics, Computer Vision, NLP, Cloud Computing, Web Applications, Image Security

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

AI/ML & Deep Learning: Python | PyTorch | Scikit-learn | Hugging Face Transformers | LoRA / PEFT | TensorFlow | Unsloth

Data Analysis & Visualization: Pandas | NumPy | Power BI | Matplotlib | Seaborn | SQL | Feature Engineering

Software Development & MLOps: Git | Linux | Docker (basic) | REST APIs | FP16/BF16 Mixed Precision | Kaggle GPU | Bash